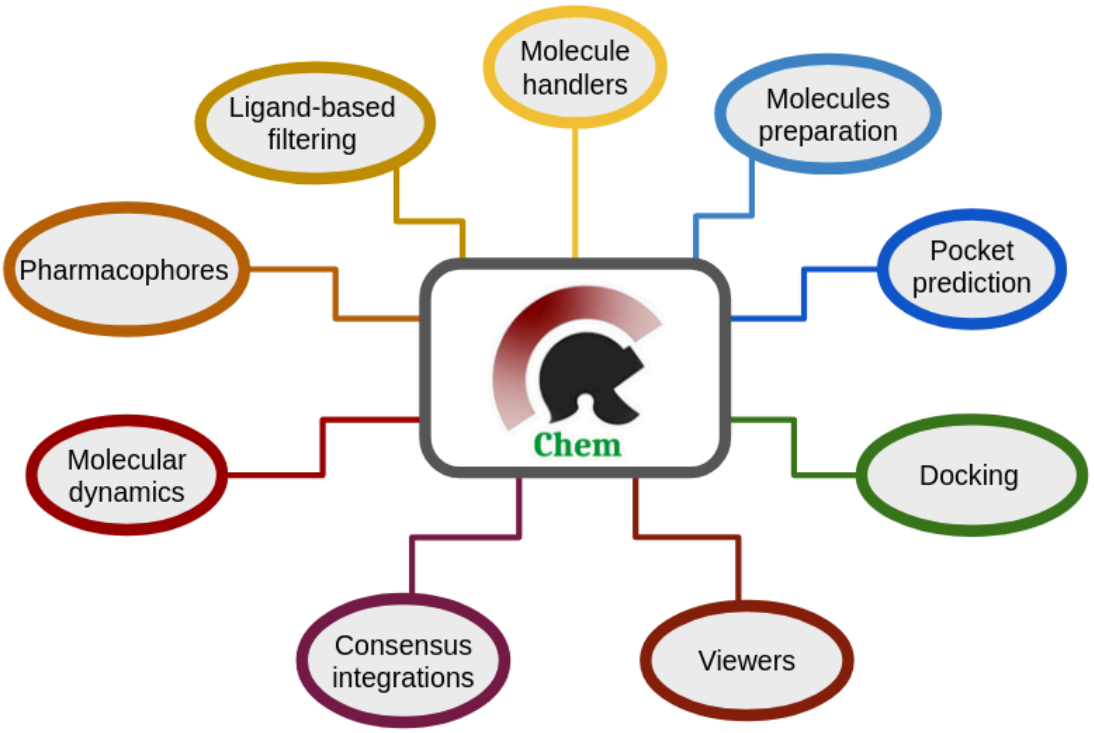
Scipion-chem Virtual Drug Screening course



This document contains a theoretical foundation for the Scipion-chem course. During this course, we will explore some examples of possible Virtual Drug Screening (VDS) workflows and tasks that can be performed on this platform. For a more hands-on guide, check the "Scipion-chem course script" file.

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Introduction to virtual drug screening

Virtual drug screening (VDS) is a computational approach used in drug discovery to identify potential therapeutic compounds from vast chemical libraries. By leveraging techniques such as molecular docking, pharmacophore modeling, and machine learning, VDS predicts how small molecules interact with target proteins, significantly accelerating the early stages of drug development. Unlike traditional high-throughput screening (HTS), which relies on physical testing of compounds, virtual screening reduces costs, time, and resource expenditure by prioritizing the most promising candidates for further experimental validation.

This method plays a pivotal role in hit identification, lead optimization, and repurposing existing drugs for new therapeutic applications. With advancements in artificial intelligence (AI) and structural bioinformatics, virtual drug screening continues to enhance precision and efficiency in modern drug discovery pipelines. However, as we will remark multiple times during the course, you must be aware that the results from all these computational analyses are predictions and therefore they need to be further tested or checked experimentally.

As illustrated in Figure 1, a standard virtual drug screening (VDS) workflow typically consists of sequential filtering stages that progressively reduce the number of candidate molecules. This stepwise reduction enables the application of increasingly sophisticated - and computationally intensive - analysis methods, which generally offer higher accuracy.

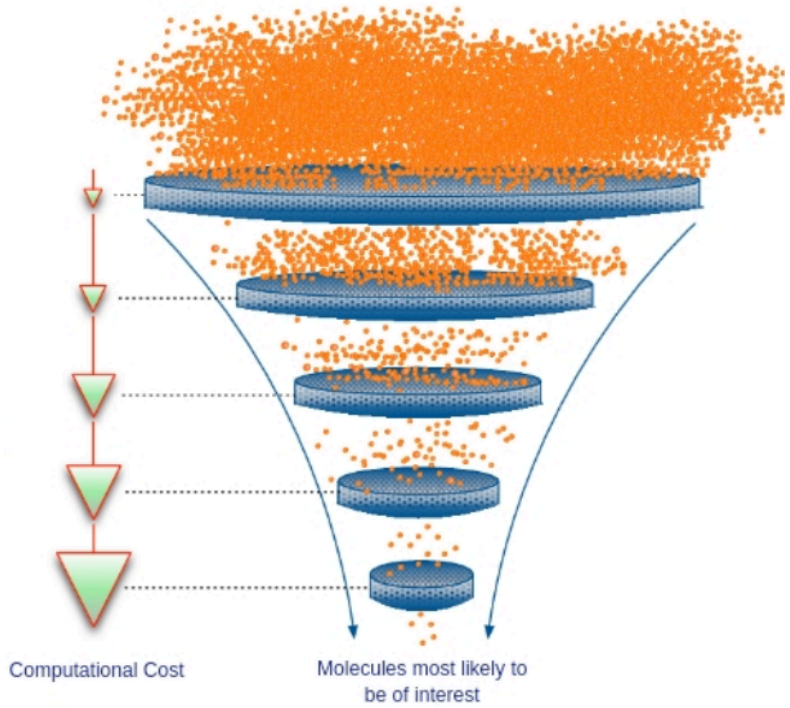


Figure 1: Standard VDS steps representation

Introduction to Scipion

Scipion is a processing framework that integrates several software packages and presents a unified interface for both biologists and developers. Scipion allows you to execute workflows combining different software tools, while taking care of formats and conversions. Additionally, all steps are tracked and can be reproduced later on.

Originally, it was specialized for generating 3D models of macromolecular complexes using electron microscopy (3DEM); however, it has since evolved with the development of **Scipion-Chem**, a dedicated branch incorporating chemoinformatics tools and logic.

Scipion basic concepts

We will now introduce some of the basic concepts of Scipion that we will be working with in the following sections:

1) **Protocol:** it stands for a single operation that can be performed inside a workflow. It usually takes one or several inputs present in the project and likewise generates one or several outputs. E.g: docking protocol.

2) **Form:** a form stands for the graphic interface where you can define all the parameters that a protocol will use for its execution.

3) **Workflow:** chain of protocols that are grouped together, connected by their inputs/outputs and that are directed to get some kind of results. E.g: simple drug screening workflow.

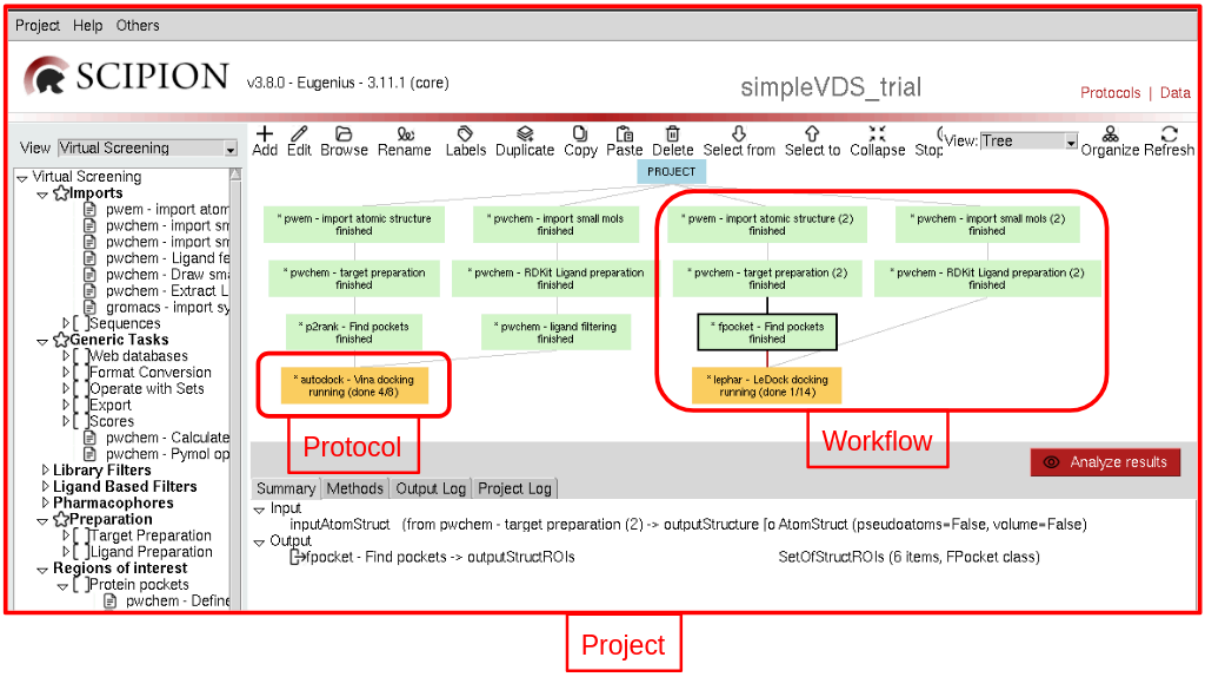


Figure 2: Basic concepts in the Scipion GUI

4) **Project**: it stands for a group of operations and workflows stored together in the same directory. Graphically, a project is displayed into a single window, that you can navigate to check the workflows and results that are present there or to create new protocols and workflows. Any new protocol can access all the outputs already present in the project.

5) **Plugin:** Scipion works with a plugin policy, where independent packages (plugins) can be installed up to the user needs. Usually, a plugin contains a single software (or group of softwares) that is useful for a certain task. E.g: scipion-chem-p2rank is the plugin that incorporates P2Rank software for the protein pocket predictions.

This plugin structure is followed in Scipion-chem. Under the Scipion core, Scipion-chem exists as a plugin, incorporating important software and managers to handle all the chemoinformatics objects; but it also functions as a dependency for the rest of the Scipion-chem plugins, which are up to the user to install.

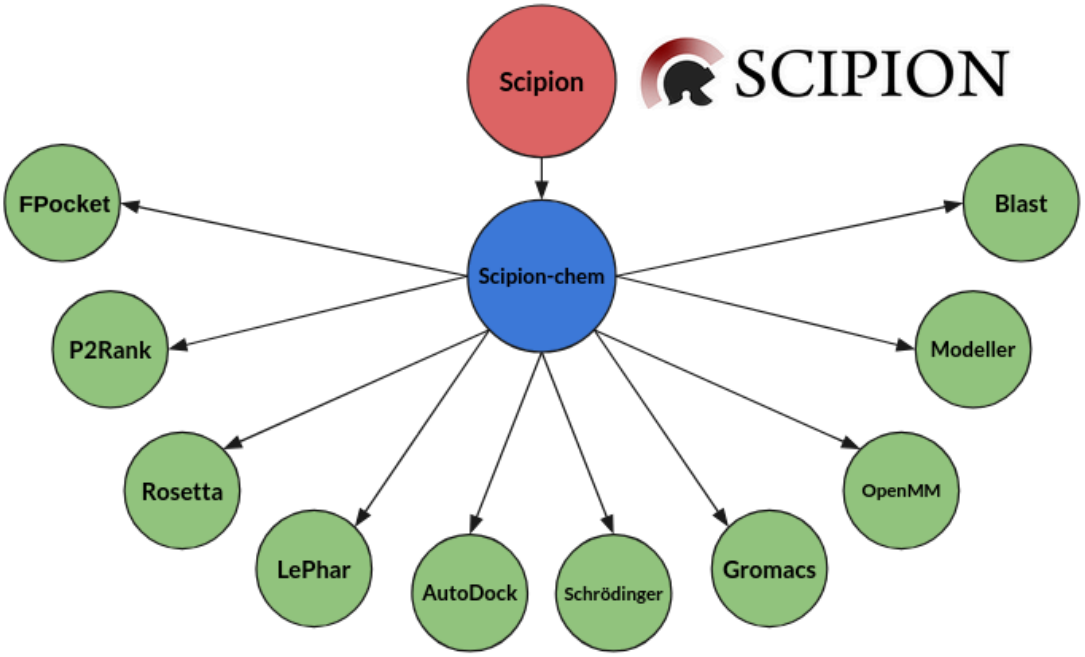


Figure 3: Scipion-chem plugin structure (not all plugins are included in the figure)

Simple VDS workflow

In this example, we show a simple VDS workflow to illustrate some of the most common steps used. We will be working with 4 small molecules and checking how their interactions with a protein receptor (PDB 4ERF) are.

0) Import elements to project

The first step for all projects is usually loading or importing some elements in our project. In our case, these are the protein receptor and the small molecules.

0.1) Import receptor: we will be importing from the PDB database the entry 4ERF.

1.2) Import small molecules: we will also be importing a set of 4 small molecules from local SDF files.

1) Prepare elements

1.1) Prepare the receptor: the structure from the PDB should be cleaned from waters, heteroatoms and, sometimes, more complex preparation. In our case, we will be just cleaning the waters, heteroatoms and keep only the chain A.

1.2) Prepare ligands: small molecules can come in a variety of formats, and sometimes some of them do not have all the 3D information needed for the structural analysis. For example, in our case, the 4 small molecules are flat, just showing us the 2D information. This protocol will generate actual conformations of these molecules, add hydrogens and in general prepare our molecules for being docked to the receptor.

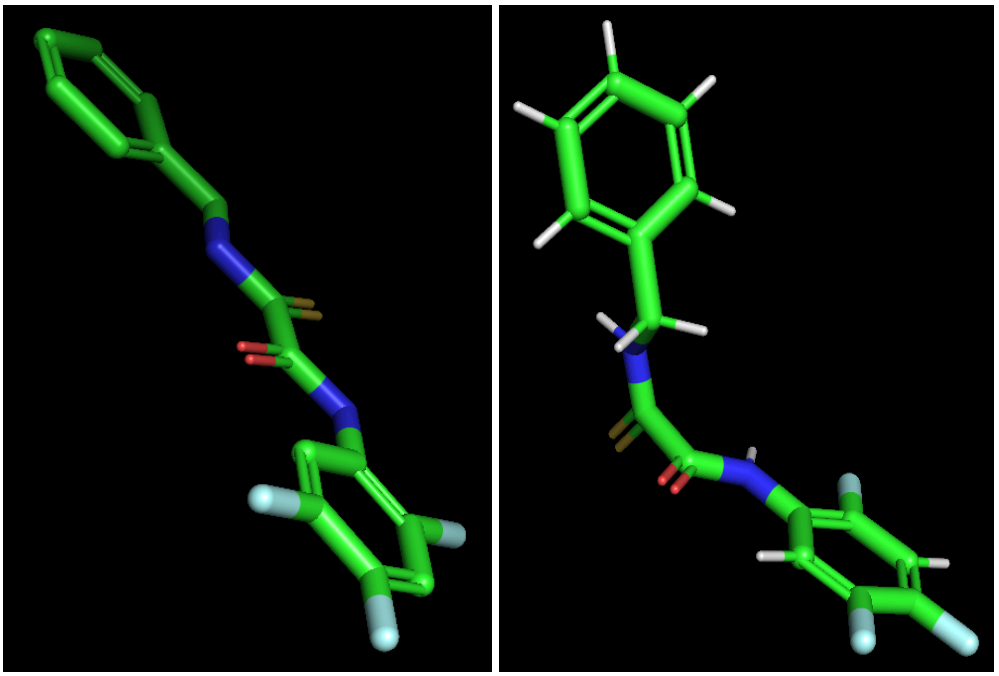


Figure 4: Small molecule "2000" in 2D (left) and 3D (right) conformations

2) Structural Regions Of Interest (ROIs)

Scipion-chem defines structural ROIs as a subset of the atomic structure residues that are of some interest for the user. Usually, in virtual drug screening workflows, these will be protein pockets, but structural ROIs are a broader definition that can include regions defined manually by the user for some other region, for example, being regions of high mutability or solvent accessibility.
In our example, from the prepared receptor, we will be launching P2Rank software to predict those sites that are more suitable to be drugged. These are usually concavities with a certain type of physicochemical characteristics. In our case, P2Rank only finds one suitable pocket. In the following steps, we will check how our molecules interact with our receptor in this particular place.

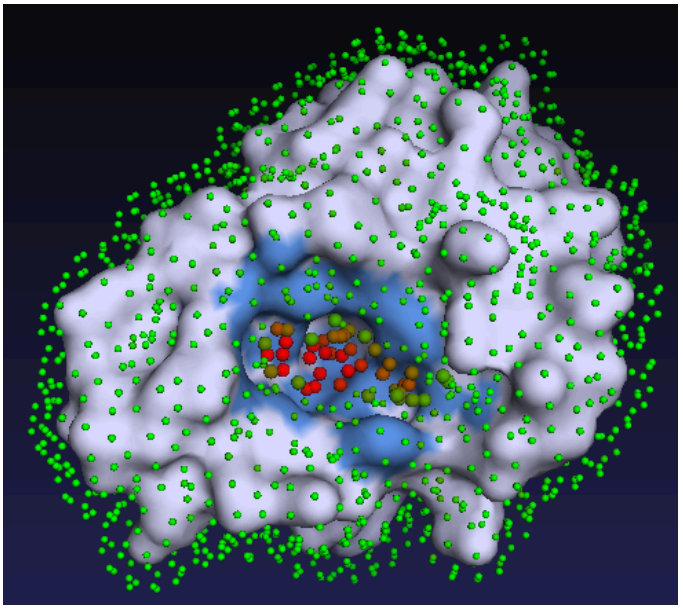


Figure 5: Representation of the P2Rank predicted pocket for 4ERF receptor

3) Ligand filtering

Different types of filters can be applied to our ligand set to thin it out and keep only those molecules more promising or able to be drugs. In our case, for example, we will be removing those molecules containing at least a Br atom, which removes molecule 2003 from our set.

4) Docking

This step tries to fit the different molecule conformations into the receptor in a way where they match physicochemically. This process can be done over the full surface of the protein, but we reduce the computational cost by directing it only to the pockets predicted in the previous analysis. Once docked, each molecule pose has a theoretical interaction energy associated, which predicts how good the receptor - ligand interaction is (the lower the energy, the stronger the interaction).

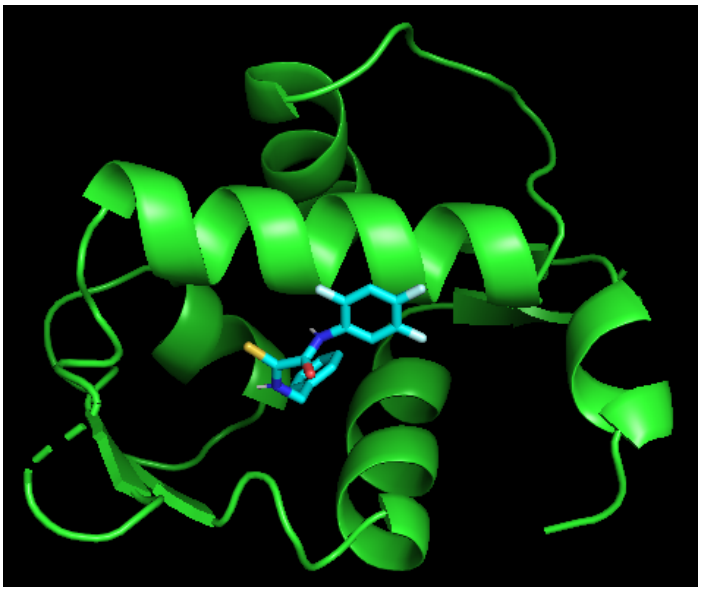


Figure 6: Molecule "2000" docked to 4ERF receptor with Vina

Analyzing PI3K selective inhibitors using Scipion-chem

PI3Ks (Phosphatidyl Inositol 3-kinases) are a family of proteins capable of phosphorylating membrane lipids (the 3′-hydroxyl group of phosphatidylinositol). This phosphorylation occurs when the associated membrane receptor is activated, generating a series of secondary messengers (PtdIns(3)P, PtdIns(3,4)P2, and/or PtdIns(3,4,5)P3 (PIP3)) that initiate a cascade of activation in the cytosol. Excessive activation of these phosphorylations is related to various types of cancer, which has prompted efforts to develop PI3K inhibitors. However, these inhibitors have typically been non-specific, binding to all (or many) isoforms of these PI3Ks, and therefore causing side effects that have prevented them from progressing to advanced stages of drug development. For this reason, the search for specific inhibitors of the different isoforms of these PI3Ks has begun, to treat more specific diseases and types of cancer. Over the past few years, molecules have been developed that have experimentally demonstrated orders of magnitude differences in binding to the various isoforms. In this study (https://doi.org/10.1038/aps.2015.71), a series of these new isoform-specific inhibitors are compiled (detailing α, δ, and providing information on others such as β, γ). In our in-silico experiment, we will attempt to retrieve specificity results for some of these selective inhibitors for the α and δ isoforms using classical structure-based virtual drug screening methods, such as docking, as well as our proprietary algorithm for aggregating results based on the score from different docking models for each ligand. The execution of these programs and the analysis of the results will be mainly carried out using the Scipion-chem platform (https://doi.org/10.1021/acs.jcim.3c01085), which contains these and other useful tools for virtual drug screening.

Materials

Input structures: PI3K isoforms α and δ structures are extracted from the **PDB** database with codes **8TDU** and **8BCY** respectively.

On the other hand, the structure of 13 small molecules listed in the cited paper as selective ligands have been downloaded from **PubChem.** The following table contains the information about its used name, PubChem codes and selective IC50 (nmol/L) values for the PI3K isoforms.

Table 1: IC50 (nmol/L) values for the PI3K selective ligands

Name	**PI3Kα**	**PI3Kδ**	**PubChem**
A66	32	1250	42636535
Acalisib	5441	12.7	11618268
Alpelisib	4.6	290	56649450
AZD6482	870	80	44137675
AZD8186	35	12	52913813
Copanlisib	0.5	0.7	135565596
Duvelisib	25	0.243	50905713
IC87114	>100000	500	9908783
Idelalisib	820	2.5	11625818
SAR260301	1539	468	135565159
Serabelisib	15	1390	70798655
Taselisib	0.29	0.12	51001932
TGX221	5000	100	9907093

Methods

In this course, we will be using the platform **Scipion-chem** (https://doi.org/10.1021/acs.jcim.3c01085) to perform an analysis on the druggability of the PI3K isoforms and we will try to recover the selectivity of the known ligands from in-silico prediction using just the input data.

Scipion-chem is an extension of the **Scipion** workflow engine (https://doi.org/10.1017/S2633903X23000132) that has been designed to integrate a series of external software and tools to ease the design and execution of chemoinformatic and drug development software. This is accomplished by (1) automatically installing the software of interest and building any necessary environment to avoid conflicts; (2) automatically converting the intermediate files using OBabel, RDKit, or Biopython when necessary; (3) offering a graphic user interface (GUI) to design and manage the workflows and running the programs; (4) saving and organizing all the parameters, workflows, and files in a straightforward folder structure; and (5) providing intuitive viewers and consensus protocols to compare and extract the most relevant information out of results from different software in equivalent steps.

The core of Scipion-chem includes several useful tools and self-made algorithms related to chemoinformatics, but then, it also integrates several plugins that install other external software the user may want to use. In this project, we will be using the following Scipion-chem plugins:

1. Scipion-chem-autodock: it includes several AutoDock tools that we will use for pocket prediction (**AutoSite**) and docking **(AutoDock-GPU and Vina)**.
2. Scipion-chem-P2Rank: it includes the software **P2Rank** for protein pocket prediction.
3. Scipion-chem-fpocket: it includes the software **FPocket** for protein pocket prediction.
4. Scipion-chem-openmm: **OpenMM** includes molecular dynamics tools used in this project to simulate the behavior of the complexes generated.
5. Scipion-chem-lephar: it includes a software for molecular docking (**LeDock**).
6. Scipion-chem-durrantlab: it includes **AutoGrow4** and **DeepFrag**, tools used in lead optimization.
7. Scipion-chem-gromacs: **Gromacs** includes software for molecular dynamics. In our case, we will be using it for studying the receptor behaviour.
8. Scipion-chem-conplex: **ConPlex** is a machine learning model that can be used to predict the affinity of huge sets of proteins and small molecules using only the sequence.

9. Scipion-chem-rosetta: Rosetta is a suite for modeling and designing biomolecules. We will use its **Flex ddG** protocol to predict how mutations change protein–protein binding energy.

10. Scipion-chem-foldxsuite: it includes the software **FoldX**, which uses a full-atom energy function to predict how mutations affect protein stability and binding.

11. Scipion-chem-saambe: it includes **SAAMBE-3D**, a machine learning method that predicts how single amino acid mutations affect protein–protein interactions.

Workflow

This project shows several functionalities included in the Scipion-chem platform in an example of PI3K isoforms and the analysis of their affinity to a series of ligands. The configuration of the steps shown in the project is just an example of how these steps or others can be combined. The tools are provided and then it is up to the user to make proper decisions in order to obtain the best results.

A **protocol**, represented as a colored box, performs a task that takes one or several inputs such as an atomic structure, that usually come from another protocol; and generates one or several outputs that can be further connected to other protocols. All protocols in the project include tools to analyze the results of their execution in the "**Analyze results**" button. These include viewers of the direct results (pockets, docking poses...) or more complex analysis.

The different stages and branches of the project are numbered and color labelled so they can be easily followed.

0) Imports and preparations

In this experiment we study and compare the PI3K isoforms alfa (PDB 8TDU) and delta (PDB 8BCY).

0.1) **Import target structures from the PDB database**

Then, we prepare them using the **Receptor Preparation** protocol: remove waters, reconstruct missing atoms, keep only the main chain and remove heteroatoms (present ligands).

0.2) Generate ligand-receptor complex structure

We keep these present ligands (X3N for alfa and I3H for delta) to study the pocket, with the same protocol.

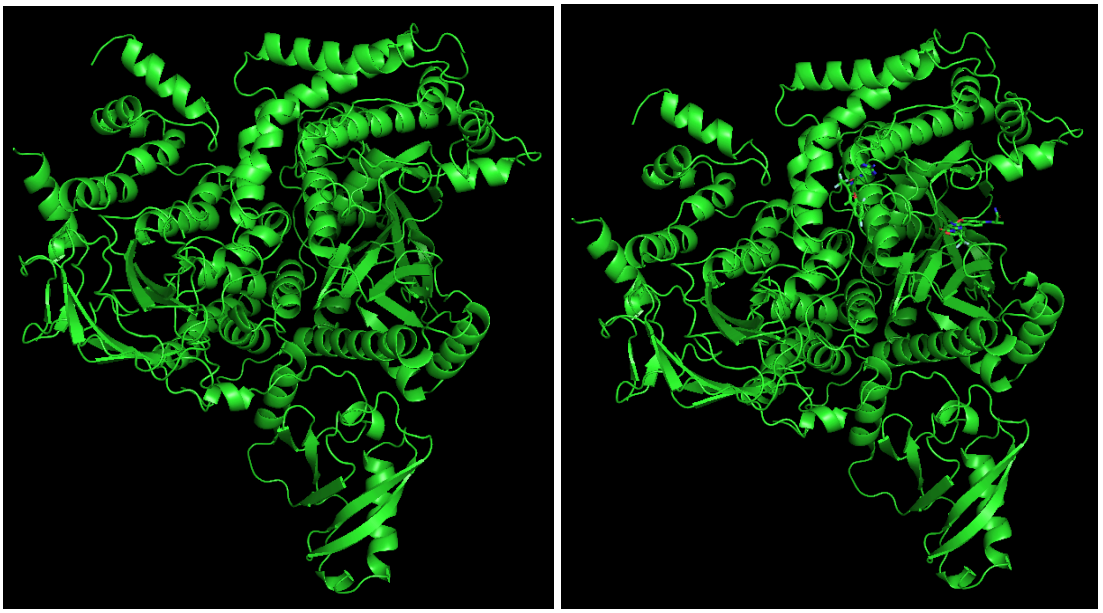


Figure 7: clean (left) and complex (right) chain A of PI3Kα. Visualization in PyMol

0.3) Import 13 known selective ligands listed in paper (https://doi.org/10.1038/aps.2015.71) from either files or **PubChem** codes and preparation and conformer generation using **RDKit.**

1) Druggable pocket analysis

We study the conformation and contacts for the pocket of the known ligand.

1.1) Pocket residues and contacts of the known ligands from the complex

From the prepared complex, we perform:

- **Ligand extraction**: ligands can then be visualized using PyMol as docked. PLIP allows us to check the interactions with the receptor.

- **Pocket extraction**: a structural Region Of Interest (ROI) is generated as the protein residues which are close to the complex ligands (X3N and I3H).

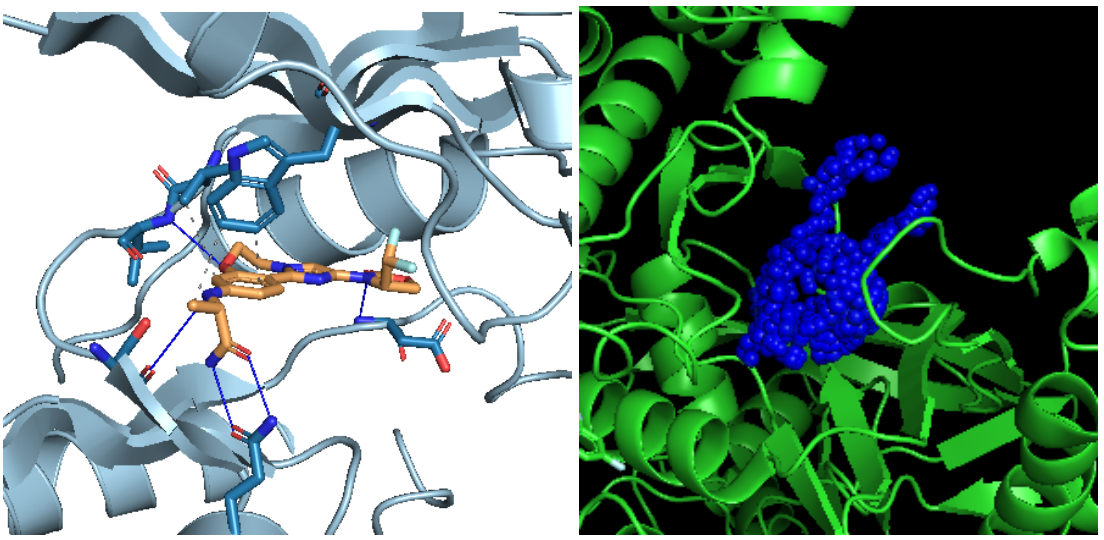


Figure 8: Contacts of the X3N ligand to PI3Ka (left) and its defined pocket (right)

1.2) Receptor and pocket attribute analysis

From the clean receptor, we can calculate attributes such as the **surface accessibility**, which can be visualized over the sequence (**AliView**) or the structure (**ChimeraX**). It can then be used to generate ROIs with the higher/lower values of those attributes over the sequence and map them back to structural ROIs.

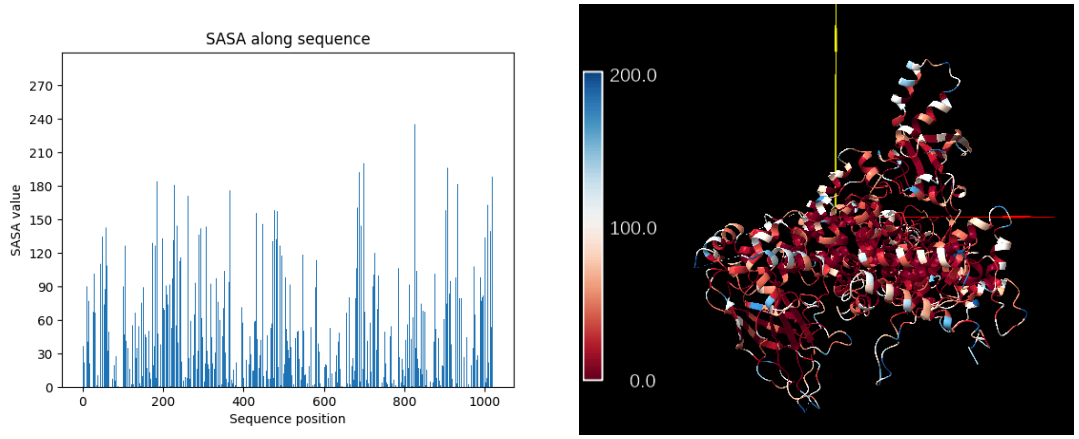


Figure 9: SASA over sequence (left) and over structure on ChimeraX (right)

1.3) Prediction of pockets and comparison to real pockets

A total of 3 different programs (**P2Rank, AutoSite** and **FPocket)** are used to predict a big number of pockets over our proteins (~50-100 each). The **consensus ROIs** protocol is able to cluster those pockets by their contact residues and return only a representative of the big enough clusters of pockets.

Performing the consensus again against the real pocket shows us that the resulting pockets indeed includes the real pocket, which supports that the real pockets are usually predicted by all the programs and shared.

However, users must be aware that the predictions might sometimes fail, so you should always check the results and act accordingly.

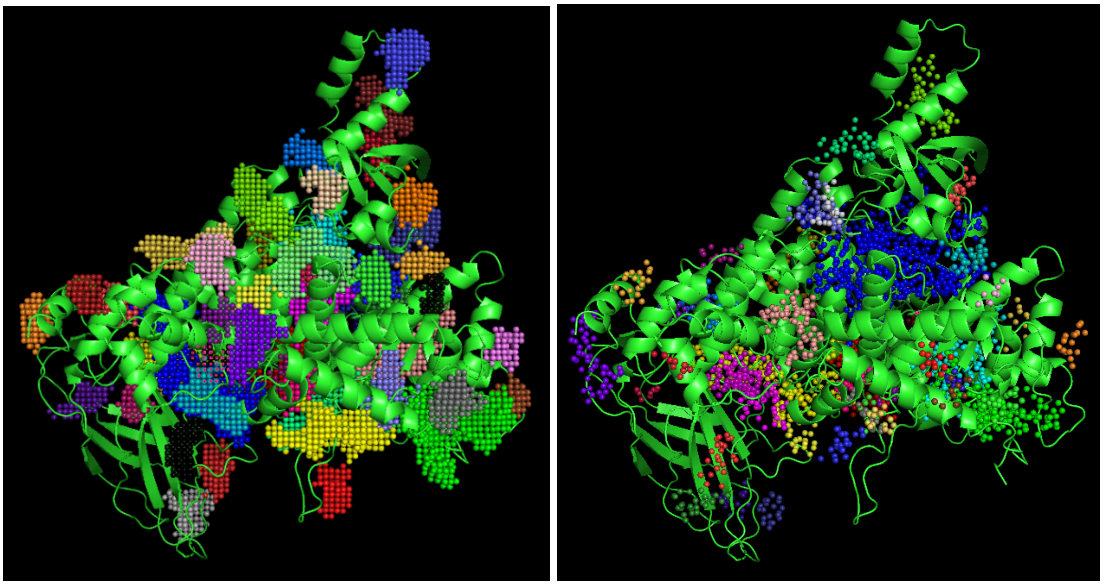


Figure 10: AutoSite (left) and P2Rank (right) predicted pockets over PI3Ka

2) Docking analysis

2.1) Docking over known pocket (to speed up) with 3 different software (**AutoDock-GPU, Vina** and **LeDock**)

With default parameters, around 150-200 poses per software are generated.

In the PI3Ka case, we show how we can filter for those with better scores for each software (best 20%) to extract those more promising.

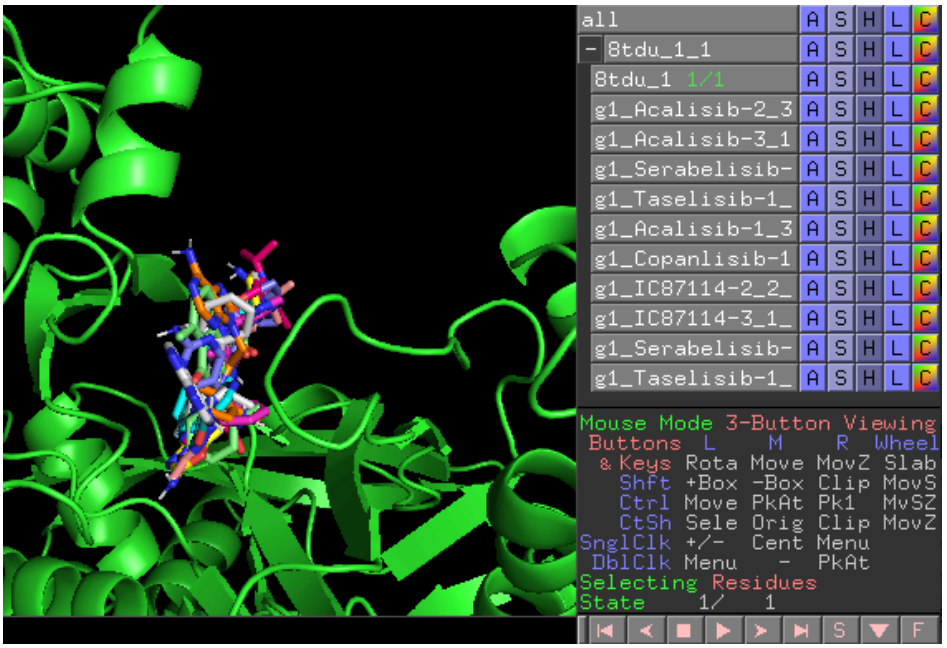


Figure 11: Poses for the consensus docking in PI3Ka

2.2) Rescoring and consensus docking to keep most promising poses

In order to make the scores of the docking poses from each software comparable, we can **rescore** the poses using **ODDT.**

The **consensus docking** is then applied to extract only those poses that are repeated among the different programs. It is made by clustering each ligand by their RMSD and keeping only a representative of the sufficiently populated clusters.

2.3) Pharmacophore generation and filtering

A set of docked ligands can be used to generate a **pharmacophore** (a 3D representation of the common physicochemical features of a set of molecules).

This pharmacophore can be then used to look for molecules that fit these pharmacophoric patterns.

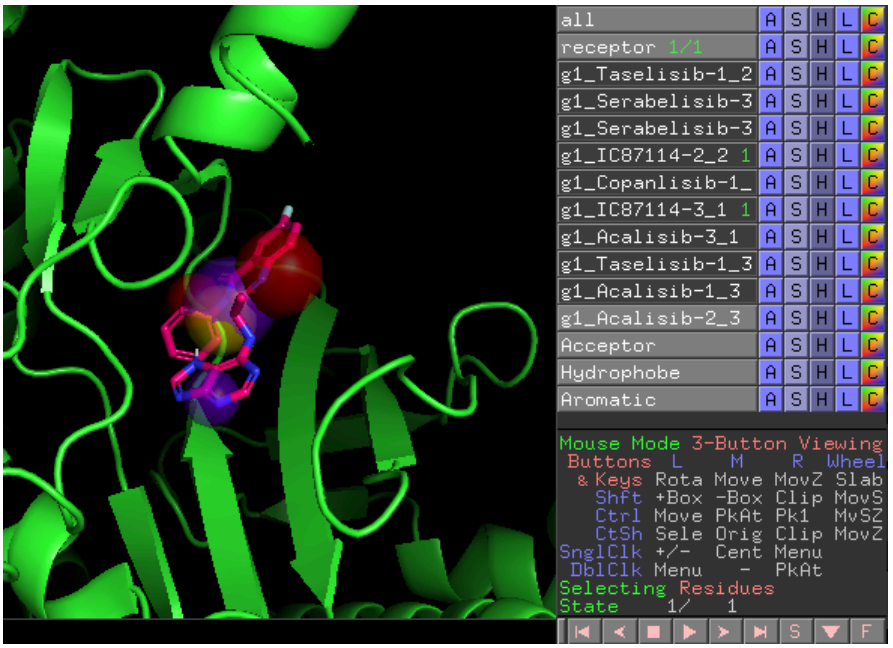


Figure 12: Pharmacophore for the consensus docking in PI3Ka

3) Docking validation

3.1) Ranking score: used to extract a score for each ligand which will depend on their position in the score ranking. It can be performed over single docking results or a combination of several.

3.2) Correlation analysis: of ranking scores and experimental log(IC50) values.

We see relatively good correlations (perfect would be -1 since IC50 are better when smaller) for most of the dockings, so best experimental ligands are usually given better scores.

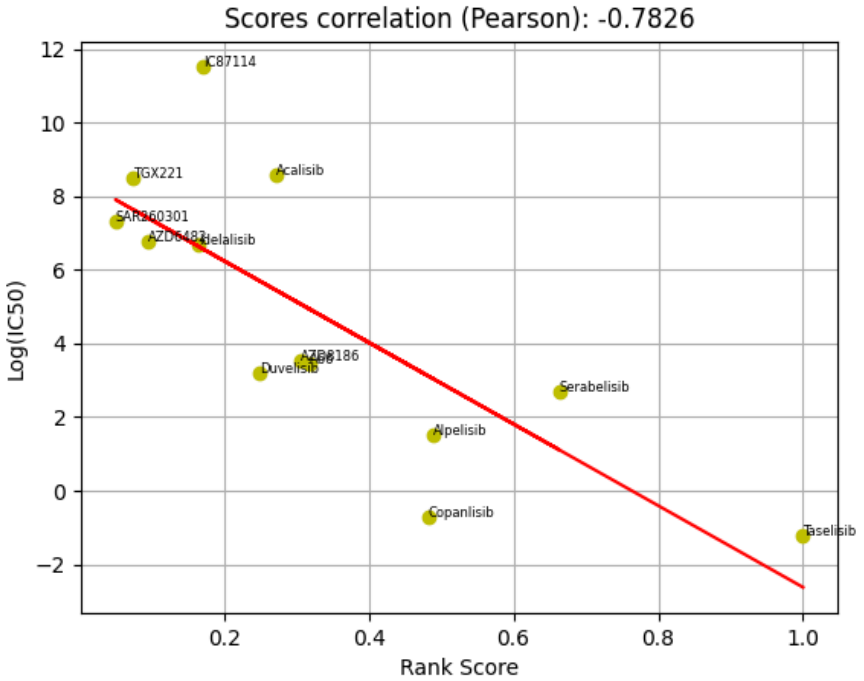
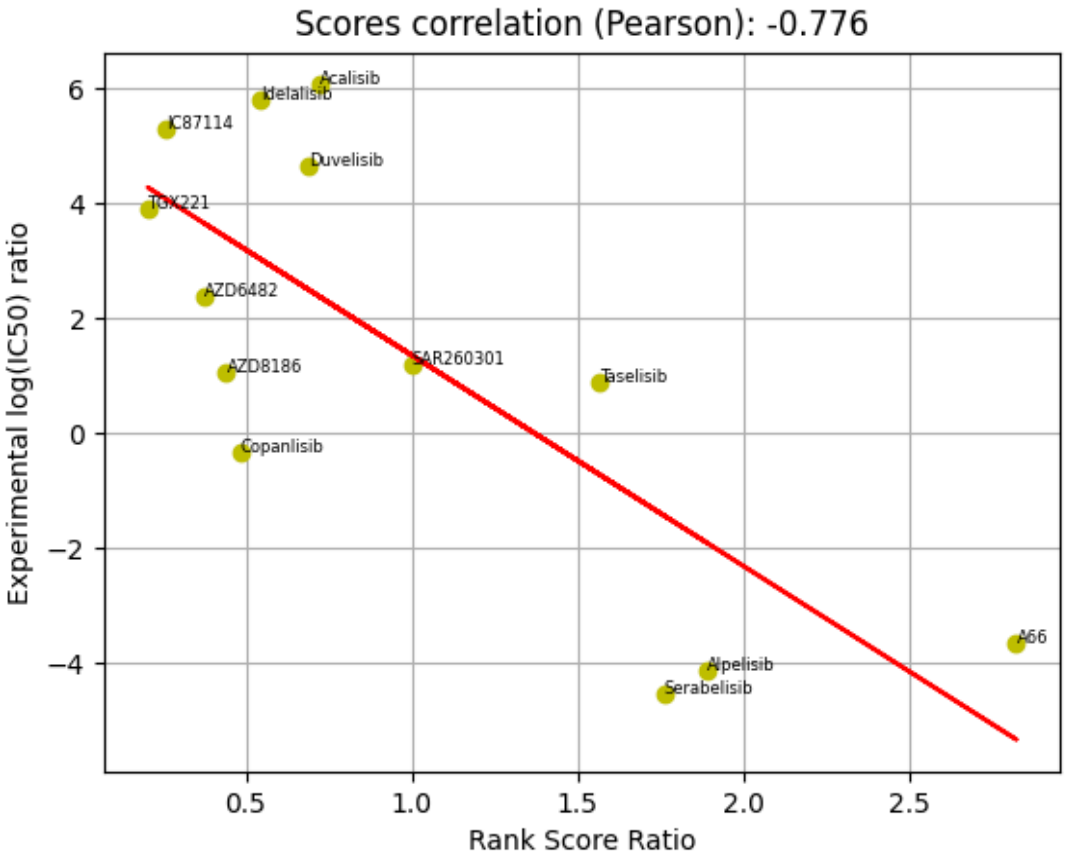


Figure 13: Correlation of rank score to actual log(IC50) scores

3.3) Correlation in the ratio alfa/delta: rankAlfa/rankDelta vs log(IC50_Alfa)-log(IC50_Delta).

This way, we study whether the specificity of the ligands in the experiment is kept by our predictions. Correlation values suggest promising results.


Scores correlation (Pearson): -0.776
Experimental log(IC50) ratio
Rank Score Ratio
Acalisib
Idelalisib
IC87114
Duvelisib
TGX221
AZD6482
AZD8186
SAR260301
Taselisib
Copanlisib
A66
Alpelisib
Serabelisib
6
4
2
0
−2
−4
0.5
1.0
1.5
2.0
2.5

Ligand filtering

Introduction to ligand filtering

Ligand filtering is an important step in virtual drug screening that reduces the number of compounds to be evaluated in subsequent analyses. Chemical libraries can contain thousands or millions of molecules, and not all of them are suitable for a particular screening campaign. Filtering allows us to select or discard compounds according to their physicochemical properties, structural characteristics, predicted biological behaviour, or similarity to known molecules.

The criteria used depend on the objective of the study. For example, some workflows prioritize compounds with physicochemical properties commonly associated with drug-like molecules, while others focus on identifying compounds structurally similar to known active ligands or matching a specific pharmacophore.

Filtering can substantially reduce the computational cost of subsequent analyses, particularly when more expensive methods such as molecular docking or molecular dynamics are planned. However, the selection criteria must be chosen carefully: excessively restrictive filters may discard potentially active compounds, including molecules that do not follow conventional drug-likeness rules.

Ligand filtering should therefore be understood as a strategy for prioritizing compounds rather than a definitive way of identifying active drugs.

Physicochemical and structural filtering

One of the simplest approaches is to filter molecules according to their chemical properties and structural characteristics.

Physicochemical descriptors include molecular weight, lipophilicity, hydrogen-bond donor and acceptor counts, and other properties that influence molecular behaviour. Structural filters can select or discard compounds based on features such as atom types, molecular size, or ring complexity.

Drug-likeness rules provide commonly used guidelines for assessing whether a compound has properties associated with established drugs. For example, Lipinski's Rule of Five considers molecular weight, lipophilicity, hydrogen-bond donors, and hydrogen-bond acceptors to identify compounds that may have an increased risk of poor oral absorption or permeability.

Other rules, such as the Rule of Three, are frequently used in fragment-based drug discovery to identify relatively small molecules suitable for further development.

These rules are guidelines rather than absolute requirements. Compounds that violate one or more criteria may still be biologically active or useful for particular applications.

ADMET filtering

ADMET stands for **a**bsorption, **d**istribution, **m**etabolism, **e**xcretion, and **t**oxicity. These properties describe different aspects of how a compound behaves within a biological system.

Computational models can estimate selected ADMET properties and help prioritize compounds with more desirable predicted profiles. For example, models may estimate intestinal absorption, blood–brain barrier permeability, metabolic stability, or potential toxicity.

ADMET filtering can be particularly useful when the objective is not simply to identify compounds that interact with a target, but also to prioritize candidates with properties that could make them suitable for further drug development.

However, these predictions have limitations. Their accuracy depends on the models, training data, and chemical space considered. Predicted properties should therefore be interpreted as supporting evidence rather than substitutes for experimental measurements.

PAINS and structural alerts

Some molecular substructures are associated with undesirable reactivity, toxicity concerns, or interference with biological assays.

Pan-assay interference compounds (PAINS) filters identify structural patterns that may be associated with interference in certain screening assays. These filters can help flag compounds that require additional investigation before experimental testing.

A PAINS alert does not automatically mean that a molecule is inactive or unsuitable as a drug. Some flagged structures may have genuine biological activity, so the results should be interpreted in the context of the target, the assay, and other available evidence.

Structural alerts can therefore be used to prioritize further inspection rather than automatically eliminating every flagged compound.

Similarity-based filtering

Another strategy is to select compounds according to their similarity to a reference molecule or a set of known active compounds.

Molecular fingerprints encode structural features of molecules into representations that can be compared using similarity or distance measures. Fingerprint-based comparison can identify compounds that share chemical features with known ligands.

Three-dimensional shape similarity compares the spatial shape of molecules. This can help identify compounds with similar three-dimensional arrangements even when their chemical structures differ.

Pharmacophore filtering selects molecules that match a set of spatially arranged chemical features considered important for molecular recognition. These features may include hydrogen-bond donors and acceptors, hydrophobic regions, and aromatic groups.

Similarity-based approaches are particularly useful when information about known active compounds is available. However, structural similarity does not guarantee similar biological activity, and structurally different compounds can sometimes interact with the same target.

Molecular clustering

Molecular clustering groups compounds according to their structural similarity. This allows a large chemical library to be organized into groups of related molecules.

Clustering can be used to reduce redundancy by selecting representative molecules from different clusters. Alternatively, it can be applied after virtual screening to retain chemical diversity among the highest-ranked candidates.

For example, when a screening campaign identifies hundreds of promising compounds, clustering can help select a smaller set that represents different chemical scaffolds rather than selecting many closely related molecules.

Clustering does not predict biological activity by itself. Its main purpose is to organize chemical space and support the selection of representative or diverse compounds.

Ligand filtering in Scipion-chem

Scipion-chem provides protocols for applying different filtering strategies to molecular libraries and sets of small molecules. Depending on the installed plugins, these include general ligand filtering, ADMET filtering, PAINS filtering, fingerprint and shape comparison, and pharmacophore-based filtering.

The general ligand filtering protocol allows molecules to be selected or discarded according to supported structural criteria, such as atom types, molecular size, and ring complexity. Other protocols provide more specialized filtering approaches.

These methods can be combined in different ways. For example, a workflow may first remove compounds that fail basic structural criteria and then apply ADMET or PAINS filtering. Similarity-based and pharmacophore filtering can be applied when suitable reference information is available.

There is no universally correct order for all filters. The choice depends on the chemical library, the screening objective, and the balance between reducing computational cost and preserving potentially interesting compounds.

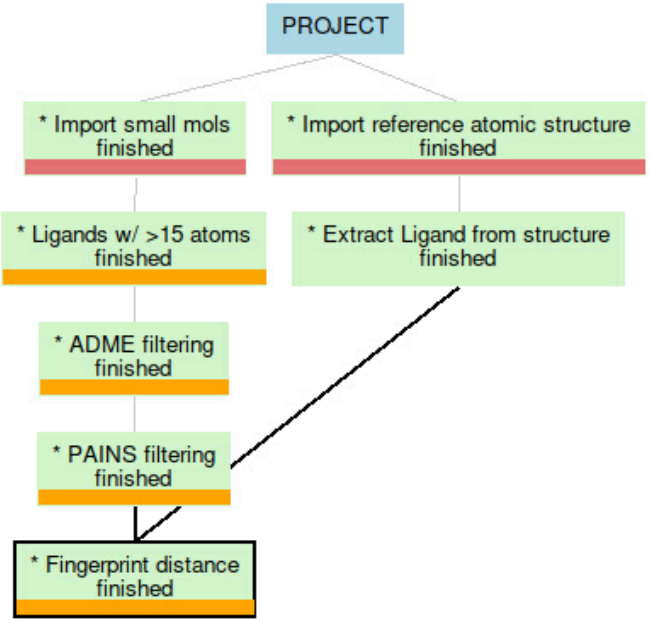


Figure 15: Example ligand-filtering workflow in Scipion-chem.

High-throughput virtual screening

Introduction to high-throughput virtual screening

High-throughput virtual screening (HTVS) aims to prioritize promising compounds from large chemical libraries against a biological target.

Traditional molecular docking predicts possible ligand-binding poses and assigns scores intended to estimate their compatibility with a target. Although docking is useful for structure-based screening, applying it to millions of compounds can require substantial computational resources.

Machine-learning-based screening methods provide alternative ways to prioritize compounds before more detailed structural analyses. These methods use representations of proteins and molecules to estimate their compatibility or rank likely candidates.

Depending on the method, the input may consist of a protein sequence, a three-dimensional protein structure or binding pocket, and a molecular representation of each candidate ligand.

HTVS does not necessarily replace molecular docking. Instead, it can be used as an initial prioritization step to reduce the number of compounds that require more computationally intensive evaluation.

Molecular clustering for large libraries

Large chemical libraries frequently contain groups of structurally similar molecules. Screening every member of these groups may introduce redundancy into the results.

Molecular clustering organizes compounds into groups based on their structural similarity. Molecular fingerprints can be used to represent compounds and calculate their similarity, allowing related molecules to be grouped together.

Representative molecules can then be selected from different clusters to reduce redundancy before screening. Alternatively, clustering can be applied after screening to retain a chemically diverse subset of the highest-ranked compounds.

For example, BitBIRCH is a clustering approach designed to organize large collections of molecular fingerprints efficiently.

Clustering is optional and should be used according to the objective of the screening campaign. If the goal is to screen every available compound, clustering may not be necessary. Furthermore, selecting only representatives can cause potentially interesting compounds to be missed.

Sequence-based screening: ConPLex

Some machine-learning methods estimate protein–ligand compatibility without requiring a three-dimensional protein structure as input.

ConPLex is a protein–ligand interaction prediction approach that uses protein sequences and molecular representations, such as SMILES, together with contrastive learning.

Contrastive learning trains a model to learn representations that distinguish compatible protein–ligand pairs from less compatible pairs. The learned representations can then be used to prioritize compounds for a particular protein target.

Because ConPLex uses protein sequence information rather than requiring an explicit three-dimensional binding pocket as input, it can be useful when a suitable target structure is unavailable.

However, predictions remain dependent on the model and its training data. A high compatibility score does not demonstrate experimental binding or establish a compound's biological activity.

Structure-based screening: OmniBind

Structure-based screening methods incorporate three-dimensional information about a target to assess its compatibility with candidate molecules.

OmniBind is an example of a machine-learning-based screening approach that uses structural information. Such methods can exploit information about the spatial arrangement of the binding site and the molecular features of candidate ligands.

The use of structural information can be valuable when a suitable target structure and binding-site representation are available. However, the quality and relevance of the structural input can influence the predictions.

As with other machine-learning screening methods, predicted compatibility scores should be interpreted according to the method's definition. They should not automatically be treated as experimental binding affinities.

Contrastive retrieval: DrugCLIP

DrugCLIP uses contrastive learning for large-scale protein-pocket–ligand retrieval and ranking.

The general principle is to learn representations of protein binding pockets and ligands in a shared embedding space. The model can then identify and rank candidate ligands according to their predicted compatibility with a target pocket.

This approach is particularly relevant to large-scale screening because it can prioritize candidate compounds without requiring a conventional docking calculation for every compound.

The resulting scores reflect the model's learned representation and ranking procedure. They are not necessarily equivalent to binding affinities measured experimentally.

Promising candidates should therefore be evaluated using additional computational approaches and, where possible, experimental assays.

Combining HTVS methods

Different HTVS approaches can be used independently or in combination.

For example, a large chemical library could first be reduced using basic physicochemical filters. One or more machine-learning methods could then rank the remaining compounds, and clustering could be used to select a chemically diverse subset of high-ranking candidates.

Alternatively, several screening methods could be applied in parallel. Their rankings could then be compared to identify compounds prioritized by multiple approaches.

The appropriate strategy depends on the size of the library, the computational resources available, the quality of the target information, and the intended use of the results.

Importantly, agreement between multiple models does not constitute experimental validation. Different methods may share assumptions, training data, or sources of bias.

HTVS in Scipion-chem

Scipion-chem provides an environment in which different computational methods can be incorporated into a common workflow. Depending on the installed plugins and their versions, users can access protocols for molecular clustering and machine-learning-based virtual screening, including ConPLex, OmniBind, and DrugCLIP.

A typical workflow begins with a chemical library. Optional filtering and clustering can reduce the number of compounds or organize the chemical space before screening.

The resulting ligand collection can then be analyzed using one or more screening methods. These methods should generally be treated as alternative or complementary approaches rather than mandatory consecutive stages.

Finally, promising candidates can be selected for more detailed analysis, such as molecular docking, inspection of predicted interactions, or experimental validation.

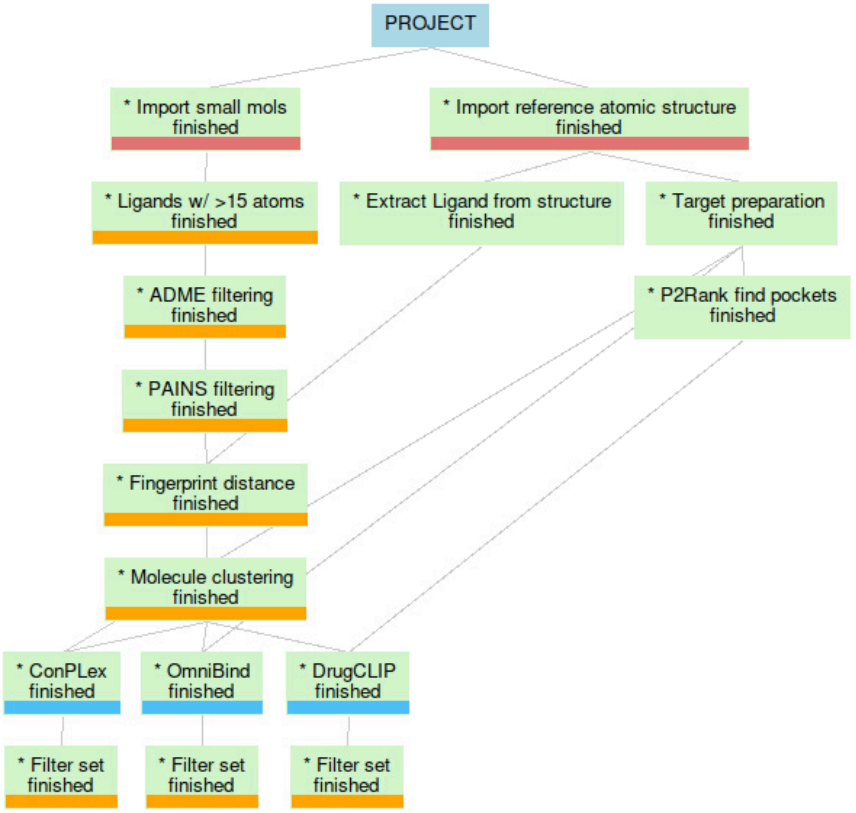


Figure 16: Example high-throughput virtual screening workflow in Scipion-chem.

Sequences Workflow

Scipion-chem includes several tools to manage and operate over protein sequences that could be useful in different bioinformatic workflows. Here, we will explore some of these tools in a simple project.

0) Import sequence

In a similar way to atomic structures, sequences (either based on nucleotides or aminoacids) can be imported to the project from a local file (in fasta sequence format or PDB protein structure format) or from typical sequence databases (Uniprot, Genebank or NCBI). Scipion-chem allows the visualization of these sequence objects using AliView.

In our example, we will, first, download a PI3Ka sequence from Uniprot, with ID: "P42336"; and second, a set of 100 sequences from the NCBI with the keyword "PI3K".

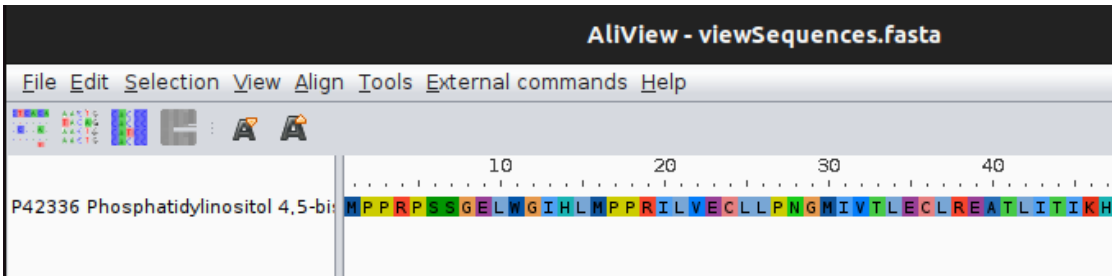


Figure 17: Protein sequence visualized in AliView

1) Sequence operations

Once the sequence has been imported in the project, we can perform a set of analysis and operations over it. In this project, we include:

- **Blast search:** the well known searcher of similar sequences. Scipion-chem allows the user to search for similar sequences in the default online databases or local databases created by the user.
- **Sequence alignment:** Scipion-chem includes 2 different protocols to align sequences: (1) pairwise alignment to align two individual sequence objects that have been imported; and (2) multiple sequence alignment to align a set of sequences object, using programs like Clustal or Muscle.

- **Structure modelling:** there are several ways to predict the structure of a protein from its sequence and Scipion includes several of these methods. In our example, we used Modeller comparative modelling, an homology based predictor, to model the structure of our PI3Ka sequence. AlphaFold is also integrated in the Scipion platform, but there is an entire modelling course in Scipion, so we will not cover that area.

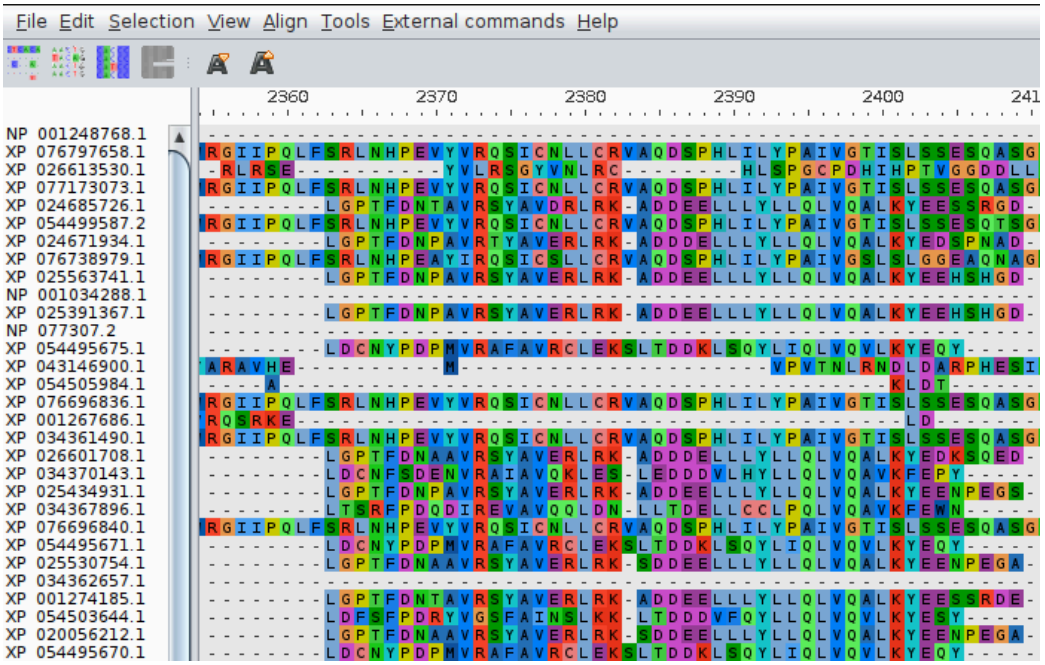


Figure 18: Multiple Sequence Alignment (MSA) of the BLAST output in AliView

2) Sequence attributes

Other protocols can be used to calculate several attributes on the sequences in the project. For example, on a set of aligned sequences, we can calculate the conservation that is mapped in our example to the consensus sequence. Another attribute that might be useful is the accessibility to the solvent, which we can calculate using the modelled PI3Ka predicted structure. These accessibility values can be mapped to both the structure and the sequence.

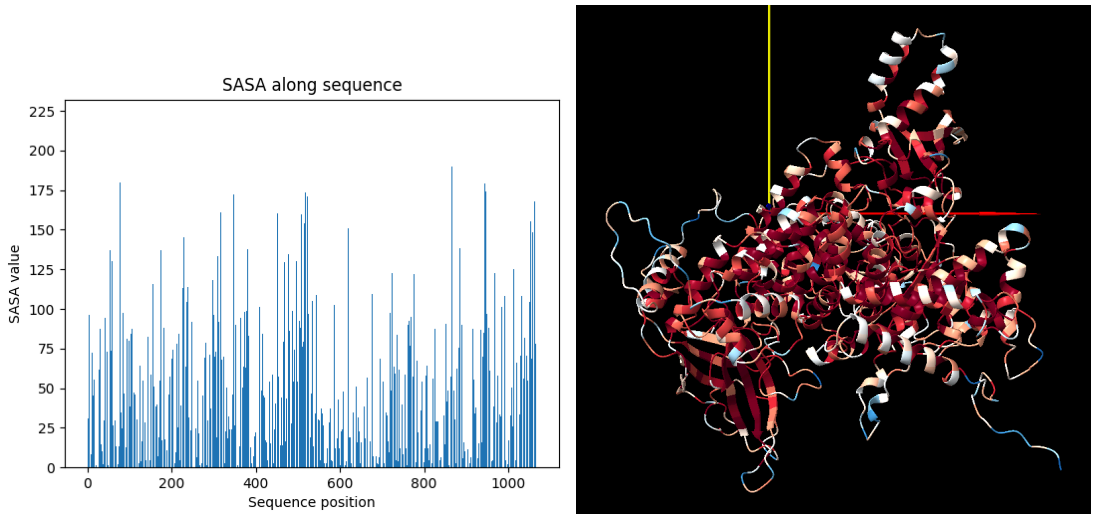


Figure 19: Representation of the solvent accessibility (SASA) over the sequence (left) and the atomic structure (right)

3) Sequence Regions Of Interest (ROIs)

Similarly to the structural ROIs we talked about in previous workflows, sequence ROIs can be defined in Scipion-chem. These can be defined through different paths:

- **Manual definition:** ROIs can be defined manually over a sequence based either on the residues positions or with a set of subsequences.
- **Extract from sequence attributes:** we can apply filters to the sequence attributes calculated previously in order to obtain regions with, for example, low conservation or high accessibility.

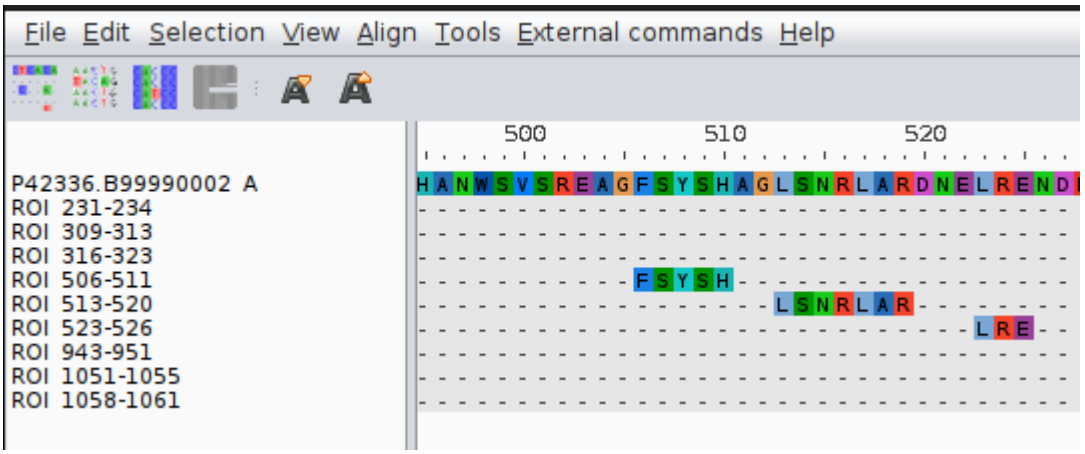


Figure 20: Sequence ROIs visualized in AliView

4) Map sequence to structural ROIs

The connection between the sequence ROIs and the structural ROIs discussed in the virtual screening workflow can be established through a mapping protocol within Scipion-chem. This protocol will take the residues in the sequence that are part of the sequence ROI, map them to an atomic structure through a pairwise alignment and then build the structural ROI over them. The user can decide whether or not to map the residue points to the closest surface of the structure.

As we mentioned before, these structural ROIs can also be defined manually with:

- The **coordinates** of the site, that will be mapped to the closest residues or surface.
- The conforming **residues** from the atomic structure.
- The **ligands** posed in the region, whose contacting residues will be the ROI.
- A **protein-protein interface (PPI)** between two chains or models.
- A **near residues pattern**, such as closeby Cys, which could be forming S-bonds.

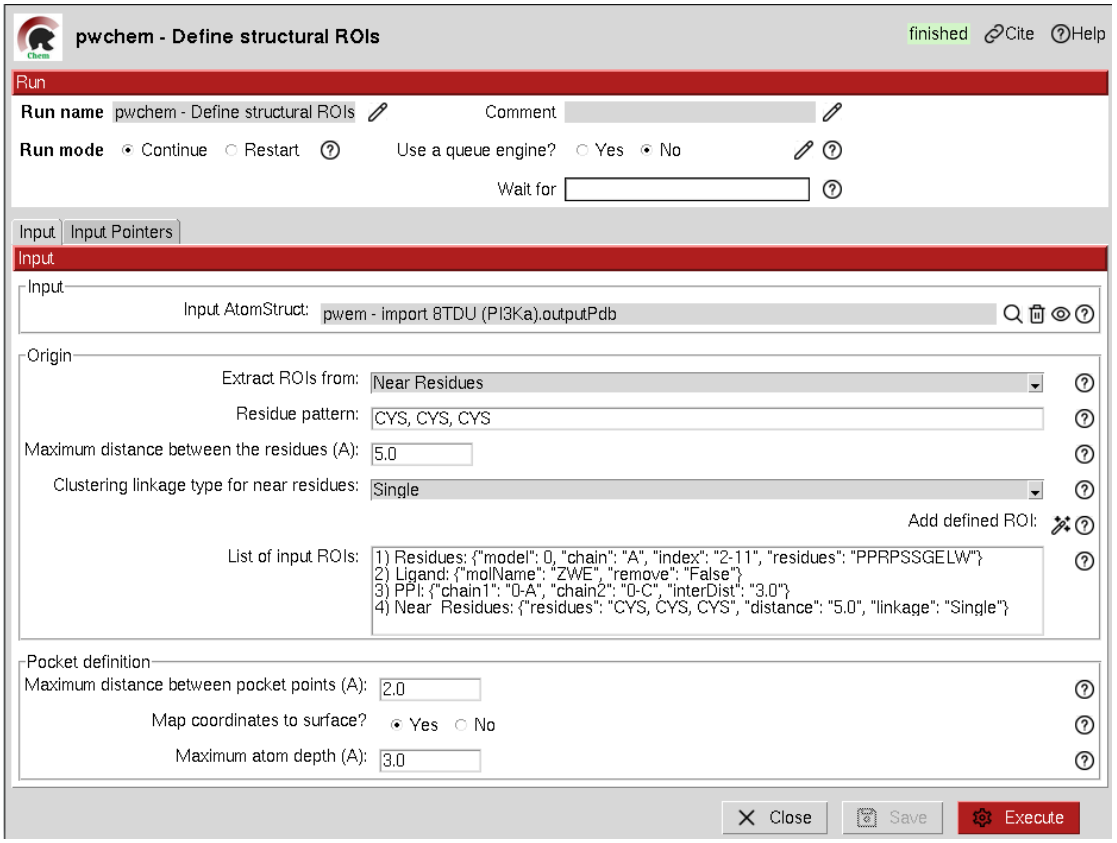


Figure 21: Form for manually defining structural ROIs

Lead Optimization

Docking allows us to identify and rank the most promising ligands for a receptor. However, these molecules are rarely optimal: once a promising hit has been found, its structure can be modified to improve its interactions with the target (**lead optimization**), and the docking itself can be refined to better represent the real behaviour of the receptor. In this section, we will explore some of the tools that Scipion-chem offers for these tasks.

1) AutoGrow4

AutoGrow4 (Scipion-chem-durrantlab plugin) is a de novo design and lead optimization tool based on a genetic algorithm. Starting from a set of seed molecules, it iteratively builds new generations of compounds, docks them into the receptor and keeps those with the best docking scores to seed the next generation. This way, the population of molecules "evolves" towards better predicted binders.

1.1) Input definition: the receptor can be defined as the whole protein or, as in our case, as a set of structural ROIs (pockets) where the docking will be focused. As seed molecules, we can use our own small molecules (for example, already known ligands, the best ligands from previous dockings) or one of the fragment libraries (from ZINC) included in AutoGrow4.

1.2) Generation of new molecules: in each generation, new compounds are created through three operators:

- Mutation: in-silico chemical reactions (from reaction libraries such as click chemistry or robust reactions) are applied to the molecules, attaching complementary fragments to them.
- Crossover: two parent molecules sharing a common substructure are combined into a new child molecule.
- Elitism: the best molecules of the previous generation are directly kept in the next one.

The new molecules are prepared with Gypsum-DL (protonation states in a pH range, tautomers, stereoisomers and 3D conformations) and docked with Vina or QuickVina2. Drug-likeness filters such as Lipinski, Ghose or PAINS can be applied during the process to avoid generating molecules with undesirable properties.

1.3) Output filtering: the output contains the docked molecules of all generations. The protocol allows keeping only those from a minimum generation on and below an energy threshold. Note that this process is computationally expensive, since each generation includes a full docking step, so the number of generations and of molecules per generation must be chosen carefully.

2) DeepFrag

DeepFrag (also included in Scipion-chem-durrantlab) is a deep learning model (a 3D convolutional neural network) trained to suggest which fragment should be added to a ligand at a certain position, given the 3D environment of the receptor pocket.

Unlike AutoGrow4, it does not explore a whole chemical space, but performs targeted modifications over a single docked ligand.

2.1) Selection of the growing point: DeepFrag works over docked poses, so the input must be a docked set of small molecules (for example, the output of the consensus docking). For each ligand, the user selects the connection point (the atom where the new fragment will be attached) and, optionally, a removal point, to replace an existing fragment of the molecule instead of just adding a new one. Scipion-chem includes wizards to visualize the ligand with its atom names and select these points.

2.2) Fragment prediction: DeepFrag returns a ranked list of the best fragments with an associated score, generating the resulting full molecules. These are produced as SMILES and can be converted into 3D structures with Gypsum-DL, OpenBabel or RDKit, so that they can be docked again to check whether the modification actually improves the interaction with the receptor.

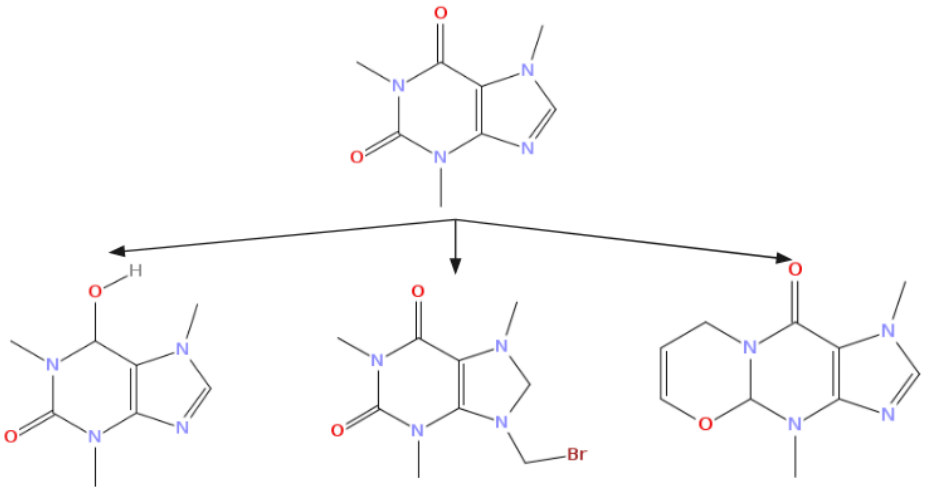


Figure 22: Examples of possible chemical variations of caffeine using DeepFrag.

3) Generative models

Inside Scipion-chem there are also 2 deep learning generative models that can propose whole new molecules for a target. Two examples available as external tools are:

3.1) REINVENT: a generative model for de novo design based on a recurrent neural network that writes molecules as SMILES strings, guided by **reinforcement learning**. A scoring function, which can combine predicted activity, docking scores, physicochemical properties and similarity to known ligands, is used to reward the generator, so that the molecules it produces drift towards the desired profile.

3.2) FLOWR.root: a more recent structure-based generative model that builds the ligand directly in 3D inside the binding pocket, instead of generating a SMILES string that then has to be docked. It uses **flow matching**, a generative technique related to diffusion models, to produce the atom types and coordinates of the molecule conditioned on the shape and chemistry of the pocket. This makes it particularly suited for lead optimization tasks such as fragment growing, linking or

scaffold hopping, where part of the ligand is kept fixed and only the rest is regenerated.

Flexible Docking

Depending on the structural flexibility of the receptor and the ligand, docking can usually be described as rigid, where both receptor and ligand are rigid structures (LeDock); semi-flexible, where receptor is rigid but ligand can have several conformations (AutoDock); and flexible, where both receptor and ligand can have some degrees of flexibility.

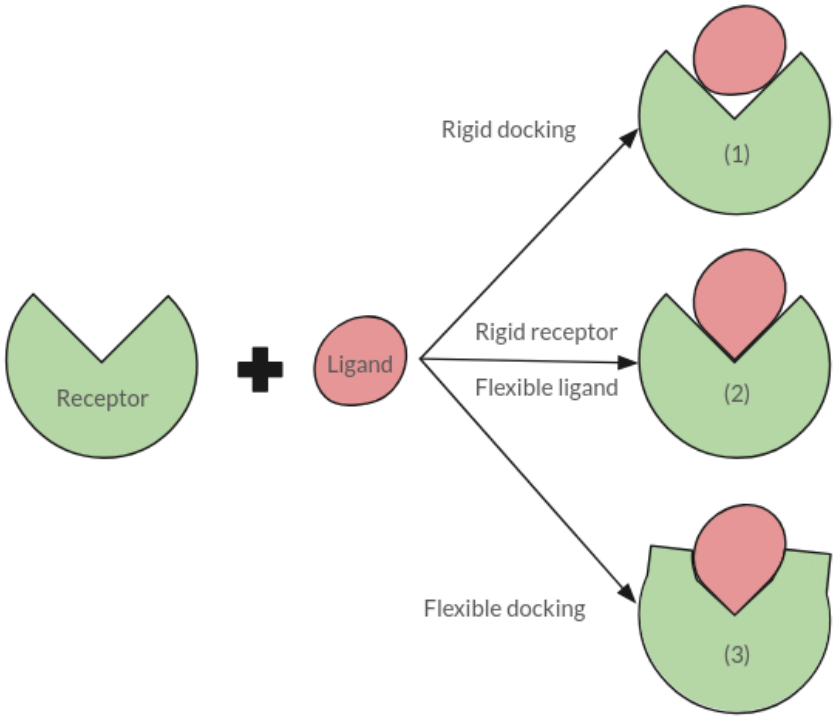


Figure 23: Scheme describing the 3 types of docking regarding flexibility

In the dockings performed so far, the receptor has been treated as a rigid body and only the ligand was flexible. However, proteins are dynamic and the side chains of the pocket residues can rearrange upon ligand binding (induced fit). Ignoring this flexibility may cause good ligands to be discarded only because they do not fit in the specific conformation of the experimental structure.

1) Docking with flexible residues

The AutoDock docking protocols in Scipion-chem (AutoDock4, Vina and AutoDock-GPU) and GNINA allow defining a set of receptor residues whose side chains will be flexible during the docking. They can be selected through wizards by chain and residue. In our case, a good choice are the pocket residues observed to interact with the known ligands (PLIP analysis in step 1.1 of the PI3K workflow).

The docking now searches simultaneously over the ligand conformations and the side-chain conformations of the flexible residues, and the output poses include the corresponding modified receptor. Since each flexible residue adds several degrees of freedom, the computational cost and the noise in the results grows quickly, so it is recommended to keep only a few residues in direct contact with the ligand as flexible. For example, AutoDock4 and Autodock-GPU sets a limit of 32 rotable bonds for ligands + flexible residues. Note that the protein backbone remains rigid: larger conformational changes are better studied with molecular dynamics.

3.2) Induced Fit Docking (Schrödinger):

The Scipion-chem-schrodinger plugin offers a different approach to receptor flexibility. Instead of sampling side-chain torsions during the docking, **Induced Fit Docking (IFD)** iterates between docking and protein refinement: the ligand is first docked with **Glide** into a softened receptor (reduced van der Waals radii), then **Prime** repacks and minimizes the residues around each pose (and optionally refines nearby loops or helices), and finally the ligand is re-docked into each refined receptor. The resulting complexes are ranked with the **IFDScore**, which combines the Glide score with the Prime energy of the receptor. Note that this plugin requires a Schrödinger license.

Molecular Dynamics

Docking provides a static picture of the receptor - ligand interaction, with a simplified scoring function and (mostly) rigid receptors. **Molecular dynamics (MD)** simulates the movement of all the atoms of a system over time by numerically integrating Newton's equations of motion, with the forces between atoms described by a force field. This allows us to study the stability of the complexes, the flexibility of the receptor and the persistence of the interactions, and to obtain better estimations of binding energies. The price is a much higher computational cost, so MD is usually applied to a handful of complexes at the end of the VDS funnel (Figure 1).

Scipion-chem includes several MD plugins that share a common MD system object, so equivalent workflows can be built with different engines. In this course we use **GROMACS** (Scipion-chem-gromacs) to study the behaviour of the receptor, and to simulate the docked complexes.

Standard MD workflow

A typical MD simulation follows these steps:

1. **Prepare structure:** clean the PDB: fix missing atoms/loops, add hydrogens, set protonation (pH), keep/curate the ligand.

2. **Parametrise:** assign the protein force field; generate ligand parameters (GAFF/CGenFF) and partial charges (AM1-BCC/RESP).
3. **Solvate & neutralise:** place the molecule in a periodic box, add explicit water (TIP3P/SPC) and add ions for charge neutrality and ionic strength.
4. **Minimise:** steepest-descent energy minimisation removes clashes and bad contacts.
5. **Equilibrate:** NVT (heat to the desired temperature) then NPT (set density/pressure); relax the solvent around the solute and prevent energy explosions.
6. **Production:** remove restraints and run the long unbiased simulation.
7. **Analysis:** extract useful information from the trajectory.

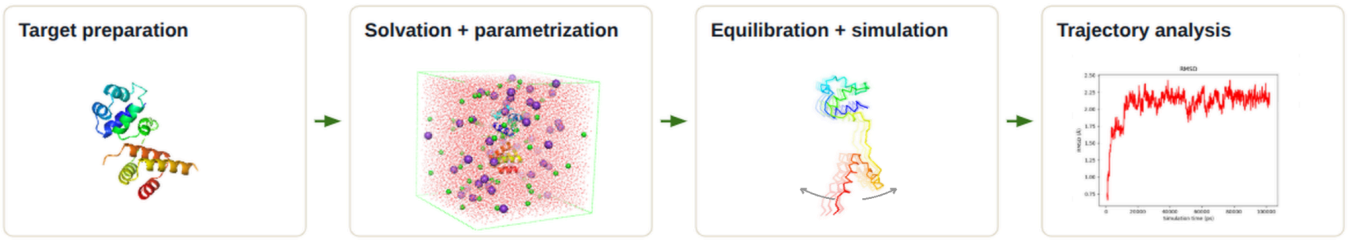


Figure 24: Scheme describing the steps of a classical MD workflow

Inside Scipion-chem, these steps are grouped into three stages:

- **System preparation protocol** (steps 1-3): prepares, parametrises and solvates the receptor or complex.
- **MD simulation protocol** (steps 4-6): runs minimisation, equilibration and production, defined as a sequence of stages.
- **Viewers and analysis protocols** (step 7): Inside scipion viewers several analysis plots can be generated like RMSD, RMSF, Energy landscape… There are additional protocols for trajectory processing and binding energy estimation and trajectory clustering.

1) System preparation

The input of an MD workflow can be an atomic structure or a docked ligand pose. In both cases, the system must be prepared before the simulation:

- **Target preparation:** Define which disulfide bridges to model, to put caps in the terminals or not and AMBER plugin also provides a membrane automatic setup.
- **Force field:** the set of parameters that describe the interactions between atoms. For proteins, common choices are Amber14 or CHARMM36. Small molecules are not included in standard protein force fields and must be parameterized separately (e.g. GAFF, OpenFF SMIRNOFF or the machine-learning based Espaloma).
- **Solvation box:** the system is placed in a periodic box filled with explicit water molecules (e.g. TIP3P), defined by a minimum distance between the solute and the box edges.

- **Ions:** counter-ions are added to neutralize the total charge of the system and, optionally, to reach a physiological salt concentration.

2) MD Simulation

The prepared system is then simulated through several consecutive stages:

- **Energy minimization:** removes steric clashes and bad contacts introduced during the preparation.
- **Equilibration:** the system is gradually brought to the target temperature (NVT ensemble) and pressure (NPT ensemble), usually restraining the heavy atoms of the solute so that the solvent can relax around it.
- **Production:** restraints are removed and the trajectory is recorded for the analysis. Its length depends on the question to answer, usually from tens to hundreds of nanoseconds.

In GROMACS, all these stages can be defined in a single protocol as a list of steps, each with its own parameters (step time, length of the simulation…). OpenMM additionally offers constant-pH simulations, where the protonation states of the titratable residues can change during the simulation.

3) Trajectory analysis

Before analysing them, trajectories are usually corrected for jumps across the periodic boundaries, centered and fitted to a reference, and cleaned from waters and ions to reduce their size. This can be done using the protocol **"Modify MDSystem"**. Then, a series of analyses can be visualized from viewers:

- **See the trajectory:** You can explore the sequential frames of the resulting trajectory using 2 viewers: PyMol and VMD. For long simulations VMD is recommended because PyMol takes a lot of memory in order to load all the frames.
- **Stability of the system**
    - **RMSD:** deviation from the initial structure over time. Its stabilization indicates that the system is equilibrated and, for the ligand, whether the docked pose remains stable in the pocket or drifts away from it.
    - **Radius of gyration and SASA:** compactness of the protein and its exposure to the solvent over time.
    - **Secondary structure analysis:** evolution of the helices, sheets and coils of each residue, showing whether the fold is kept or partially lost during the simulation.
- **Flexibility and conformational changes**
    - **RMSF:** flexibility of each residue along the simulation, highlighting mobile regions such as loops.
    - **Principal Component Analysis (PCA):** extracts the main collective motions of the protein from the atomic fluctuations, separating large functional movements (e.g. domain or loop opening) from local noise.
    - **Free energy landscape:** projection of the trajectory onto two variables (e.g. the first two principal components, or RMSD and radius of

gyration), converted into free energy. Its minima correspond to the most stable conformational states visited by the system.
    - **Clustering:** the protocol "Trajectory Clustering" groups similar frames to extract representative conformations of the receptor, which can be used, for example, as targets for new docking campaigns (ensemble docking).
- **Protein - ligand interactions**
    - **Minimum distance:** smallest distance between the ligand and the receptor (or its pocket residues) over time, a simple indicator of whether the ligand stays bound or leaves the pocket.
    - **Hydrogen bonds:** number and persistence of the polar interactions between the ligand and the receptor.
    - **Interaction analysis:** the protocols "ProLIF analysis" and OpenMMDL (only in OpenMM plugin) extract the interactions along the trajectory, showing the persistence of each contact. Interactions maintained during most of the simulation are more likely to be relevant than those only present in the docked pose.

Atomic modeling

Introduction to atomic modeling

Atomic modelling is the process of generating, constructing, or refining three-dimensional models of biomolecules at atomic resolution. These models provide information about the spatial arrangement of atoms and can be used to investigate molecular structure, interactions, and biological function.

There are different ways to obtain atomic structures. Experimental techniques such as X-ray crystallography and cryo-electron microscopy provide structural information that can be used to build or refine atomic models. Computational methods can also predict protein structures from amino acid sequences or generate models of biomolecular complexes.

These approaches are particularly useful when an experimentally determined structure is unavailable or when additional structural hypotheses are required to investigate a biological question.

Atomic modelling can also support drug-discovery workflows. For example, a protein structure can be used to investigate potential binding sites and perform molecular docking. Predicted structures of biomolecular complexes can provide hypotheses about possible molecular interactions.

However, the reliability of a model depends on the method, the available input information, and the molecular system being studied. A predicted structure should not automatically be considered equivalent to an experimentally determined structure.

Structure prediction, refinement and evaluation

Three related tasks should be distinguished.

Structure prediction involves generating a structural model from available information, such as an amino acid sequence or a description of the components of a biomolecular complex.

Model refinement aims to improve an existing structural model. Depending on the method, refinement may involve optimizing atomic geometry, adjusting the structure to experimental data, or improving the fit between a model and an experimental map.

Model evaluation involves assessing the quality and reliability of a model. This may include examining confidence estimates, checking structural geometry, identifying potential clashes, and comparing the model with experimental observations.

These tasks are complementary, but they are not interchangeable. A structure-prediction method does not necessarily refine an existing model, and a model that has been refined is not necessarily correct.

When experimental data are available, they provide an important source of evidence for assessing whether a model adequately represents the underlying molecular structure.

BioFold: integrated atomic modeling in Scipion

BioFold is a Scipion plugin that integrates several biomolecular structure-prediction methods into a common workflow environment.

The methods available in the plugin include AlphaFold 3, Protenix, IntelliFold, Boltz-2, and Chai-1, depending on the installed versions and supported protocols.

Integrating several methods into Scipion allows users to incorporate structure prediction into broader workflows involving molecular preparation, structural analysis, and downstream applications.

Depending on the selected method, users can provide the required molecular inputs, run supported methods locally, or import structures generated using external prediction servers.

The input requirements and capabilities differ between methods. Consequently, users should select a method according to the molecular system they want to model and the information available.

Overview of the available methods

AlphaFold 3

AlphaFold 3 predicts the structures of biomolecular complexes, including systems involving proteins, nucleic acids, small molecules, ions, and other supported molecular components.

Its ability to model different types of molecular interactions makes it relevant to a wide range of structural-biology questions.

Protenix

Protenix is an open-source approach to biomolecular structure prediction inspired by AlphaFold 3. It provides an alternative for predicting structures of supported biomolecular systems.

IntelliFold

IntelliFold is another deep-learning-based approach to biomolecular structure prediction. It provides an additional option for generating structural models within the integrated workflow.

Boltz-2

Boltz-2 combines biomolecular structure-prediction capabilities with binding-affinity-related predictions for supported interactions. These capabilities

make it relevant to applications involving protein–ligand interactions and computational drug discovery.

Binding-affinity predictions are computational estimates and should not be treated as equivalent to experimental affinity measurements.

Chai-1

Chai-1 is a multimodal biomolecular structure-prediction method. Its approach allows different types of biological information to be incorporated into structure prediction, with support for additional constraints in appropriate settings.

The precise inputs and capabilities available through Scipion depend on the integrated implementation and protocol.

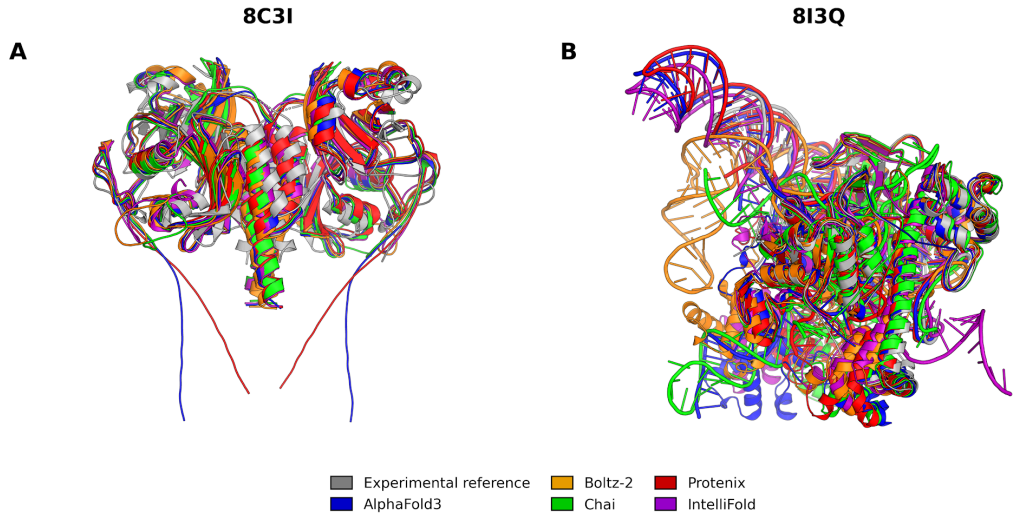


Figure 25: Examples of structures predicted using different methods.

When presenting a comparison, it is important to distinguish the methods' supported inputs and outputs rather than assuming that every method has identical capabilities or performs equally well on all molecular systems.

Atomic modeling workflow

A typical atomic modelling workflow can be divided into four main stages.

1. Define the modelling task

First, determine what needs to be modelled. The objective could be to predict the structure of an individual protein, investigate a protein complex, or model a system containing nucleic acids or other molecular components.

2. Select the method

The appropriate method depends on the molecular system, the required inputs, the available experimental information, and the computational resources. Some methods can be run locally, while others may require access to external prediction servers.

3. Generate or import the model

The selected method can be used to generate a structural model. Alternatively, a structure generated externally can be imported into Scipion for further analysis.

4. Inspect and evaluate the model

Once a model has been generated or imported, its quality should be assessed. This can involve examining confidence estimates, checking structural geometry, and identifying potential clashes.

When experimental data are available, the model can also be evaluated against those observations. For example, a model derived from cryo-EM data can be assessed in relation to the experimental density map using appropriate validation tools.

The resulting model can then be used in downstream analyses, such as studying molecular interactions, investigating binding sites, or performing molecular docking.

Generating a structure is therefore only one part of the process: its reliability must be assessed in relation to the intended application.

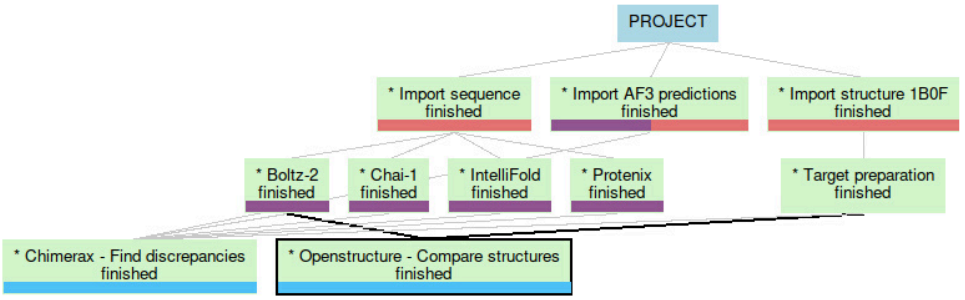


Figure 26: General atomic modeling methods comparison workflow.

Atomic modeling and virtual drug screening

Atomic modelling and virtual drug screening are closely related but address different questions.

Atomic modelling focuses on generating or evaluating structural representations of biomolecules. Virtual drug screening focuses on identifying and prioritizing candidate compounds that may interact with a biological target.

A predicted protein structure may provide a useful starting point for screening when an appropriate experimental structure is unavailable. However, the reliability of subsequent docking or screening calculations depends partly on whether the predicted structure adequately represents the relevant binding site.

For example, a prediction may reproduce the overall protein fold but still have inaccuracies in a binding pocket or in the orientation of particular side chains. These differences can influence the predicted poses and scores of candidate ligands.

It is therefore important to assess the suitability of a structure for the intended application rather than relying solely on the fact that a prediction method has produced a model.

Protein Protein Interaction

In silico mutagenesis and binding free energy changes

Introduction

Once the interface has been characterised, a natural next step is to ask how much each residue contributes to binding. Computationally, this is done by introducing point mutations in the structure of the complex and estimating the change in binding free energy:

$$\Delta\Delta Gbind = \Delta G_{bindmutant} - \Delta G_{bindwild\text{-}type} \quad (1)$$

With this convention, $\Delta\Delta G > 0$ indicates that the mutation weakens the interaction. The most common application is computational alanine scanning, in which every interface residue is mutated to alanine, removing the side chain beyond Cβ. Residues whose mutation produces $\Delta\Delta G \geq 2$ kcal $mol-1$ are usually considered hotspots.

Other strategies include saturation mutagenesis, in which all 20 amino acids are tested at a given position, and targeted scanning, in which only selected positions or substitutions are evaluated (for example, residues of known functional relevance, natural variants or mutations previously tested experimentally). Targeted scanning reduces the number of calculations and therefore allows the use of more demanding methods.

The tools below differ mainly in how they estimate $\Delta\Delta G$: an empirical force field, a machine-learning model, or a conformational ensemble.

Overview of the available tools

FoldX

FoldX was developed in Luis Serrano's group (originally at EMBL Heidelberg, now at the Centre for Genomic Regulation, CRG, Barcelona) in collaboration with Rousseau and Schymkowitz (VIB–KU Leuven). It is an empirical force field: the free energy is a weighted sum of terms (van der Waals, solvation, hydrogen bonds, electrostatics, entropy, …) fitted against experimental stability data. Mutations are modelled on a fixed backbone with local side-chain repacking, which makes FoldX very fast.

Typical workflow: RepairPDB (fixes clashes and poor rotamers), BuildModel (introduces mutations) and AnalyseComplex (interaction energy), with AlaScan and PositionScan for scanning. The latest version adds π-stacking, extends pH dependency to all charged residues and recalibrates hydrogen bonds and solvation.

Resolution dependence. Because FoldX evaluates a single, fixed-backbone structure, it is sensitive to structural quality. Its stability dataset was built from X-ray structures with resolution < 2.5 Å, and small steric clashes in lower-resolution structures can propagate into large FoldX scores. In this work, FoldX values from structures with resolution above 2.8 Å are given less weight in the z-score computation.

SAAMBE-3D

SAAMBE-3D (Single Amino Acid Mutation Binding free Energy change, 3D) was developed by Pahari et al. in Emil Alexov's group (Clemson University, USA). It is a machine-learning approach: a gradient-boosting model trained on experimental data from SKEMPI 2.0 using features extracted from the 3D structure of the complex and the physico-chemical properties of the wild-type and mutant residues. No structural model of the mutant is built, so predictions are almost instantaneous and suitable for large-scale scans.

It offers two modes: ΔΔG prediction (Pearson r ≈ 0.8 in cross-validation on SKEMPI 2.0) and classification of mutations as disruptive or non-disruptive. It is limited to single-point mutations.

Flex ddG

Flex ddG was developed by Barlow et al. in Tanja Kortemme's lab (University of California, San Francisco). It is a Rosetta-based, ensemble protocol: starting from the complex, it generates an ensemble of conformations around the mutation site with backrub moves (local backbone flexibility), followed by side-chain repacking and minimisation of both the wild-type and mutant complexes. ΔΔG is averaged over the ensemble (typically ∼35 independent runs).

Modelling backbone flexibility improves predictions, especially for small-to-large mutations, at the cost of a much higher computational time and requiring a Rosetta installation.

Comparison

Table 1 summarises the three approaches.

Table 1. Comparison of tools for predicting mutational changes in binding free energy.

Tool	Developer	Basis	Structural flexibility	Cost	Best suited for

FoldX (Schymkowitz et al., 2005; Delgado et al., 2025)	CRG Barcelona (Serrano) & VIB–KU Leuven	Empirical force field fitted to experimental data	Fixed backbone, side-chain repacking	Seconds per mutation	Alanine/position scanning, stability and binding
SAAMBE-3D (Pahari et al., 2020)	Clemson University (Alexov), 2020	Machine learning (gradient boosting) on 3D-structure features, trained on SKEMPI 2.0	None (no mutant model is built)	Near-instant	Large-scale screening of single mutations
Flex ddG (Barlow et al., 2018)	UCSF (Kortemme), 2018	Rosetta energy function averaged over a conformational ensemble	Backrub backbone sampling + repacking + minimisation	Hours per mutation (CPU)	Accurate ΔΔG for selected mutations

In practice, these tools are complementary: SAAMBE-3D and FoldX can be used to scan many positions quickly, and Flex ddG to refine the most interesting candidates. Combining their predictions (e.g. through z-scores) reduces the bias of any single method.

In silico mutagenesis workflow

A typical in silico mutagenesis workflow can be divided into four main stages.

1. Obtain the protein complex

The starting point is the 3D structure of the protein complex, which can be:

Experimental: a structure determined experimentally and downloaded from the PDB. Its resolution should be checked (see FoldX, resolution dependence).

Predicted: when the complex has not been characterised experimentally, it can be predicted. In Scipion-chem this can be done with Boltz-2 and Chai-1, or AlphaFold3 predictions can be imported. The confidence of the interface (e.g. ipTM) should be checked before continuing.

2. Define the interface

Next, the residues at the protein–protein interface are identified. These will be the positions to mutate.

ROI: residues of one chain within a distance threshold (in Å) of the partner chain are selected.

COCaDA: based on the COCaDA web server, interface residues are identified from the interactions they form (e.g. hydrogen bonds, salt bridges, hydrophobic contacts), not only from distance.

3. Calculate ΔΔG

The interface positions are mutated (saturation mutagenesis) and ΔΔG of binding is estimated with one or more methods:

FoldX: empirical force field on a fixed backbone; fast.

SAAMBE-3D: machine-learning model; near-instant, no mutant structure is built.

Flex ddG: Rosetta conformational ensemble; most accurate but computationally expensive.

4. Build a consensus

Because each method has its own scale and biases, the ΔΔG values are normalized (zmuv) and fused (median) into a consensus score, which is used to rank the mutations and identify hotspots.

No single ΔΔG prediction should be taken as definitive: agreement between methods increases confidence in the results, which can then guide experimental validation or further design.